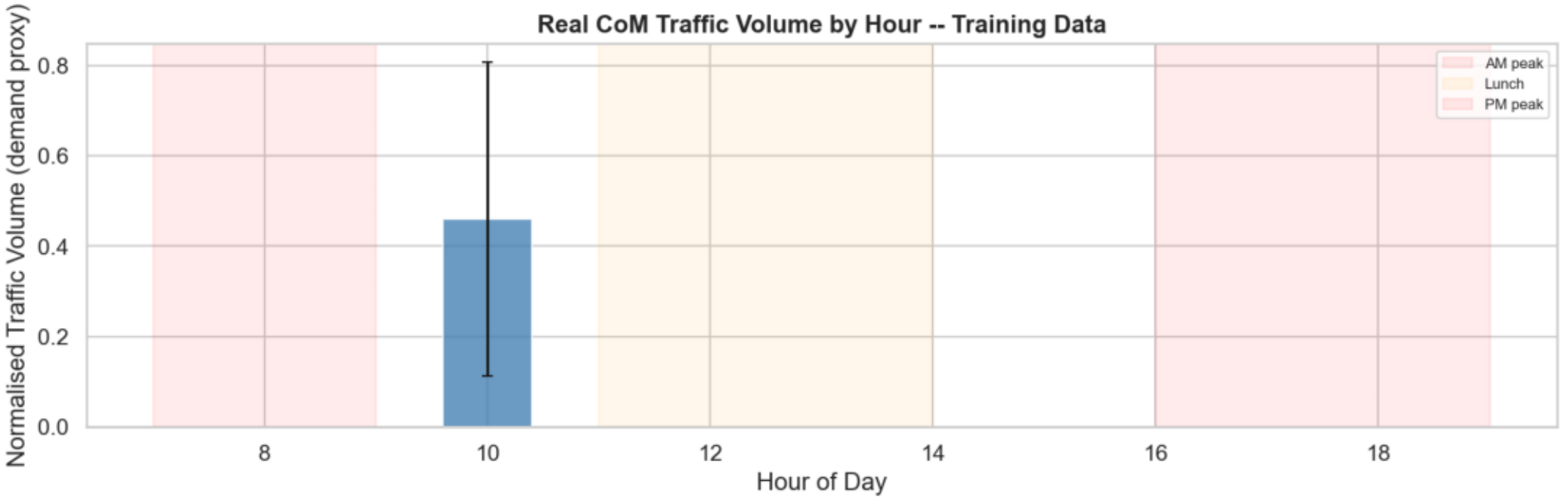
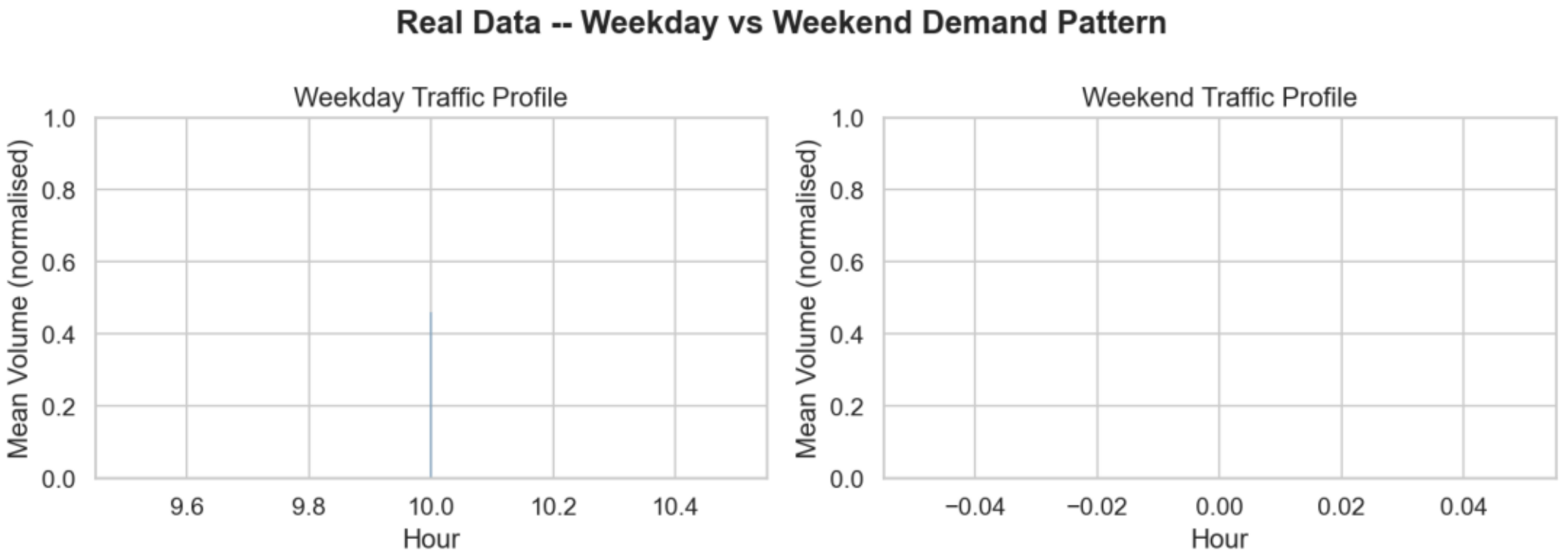
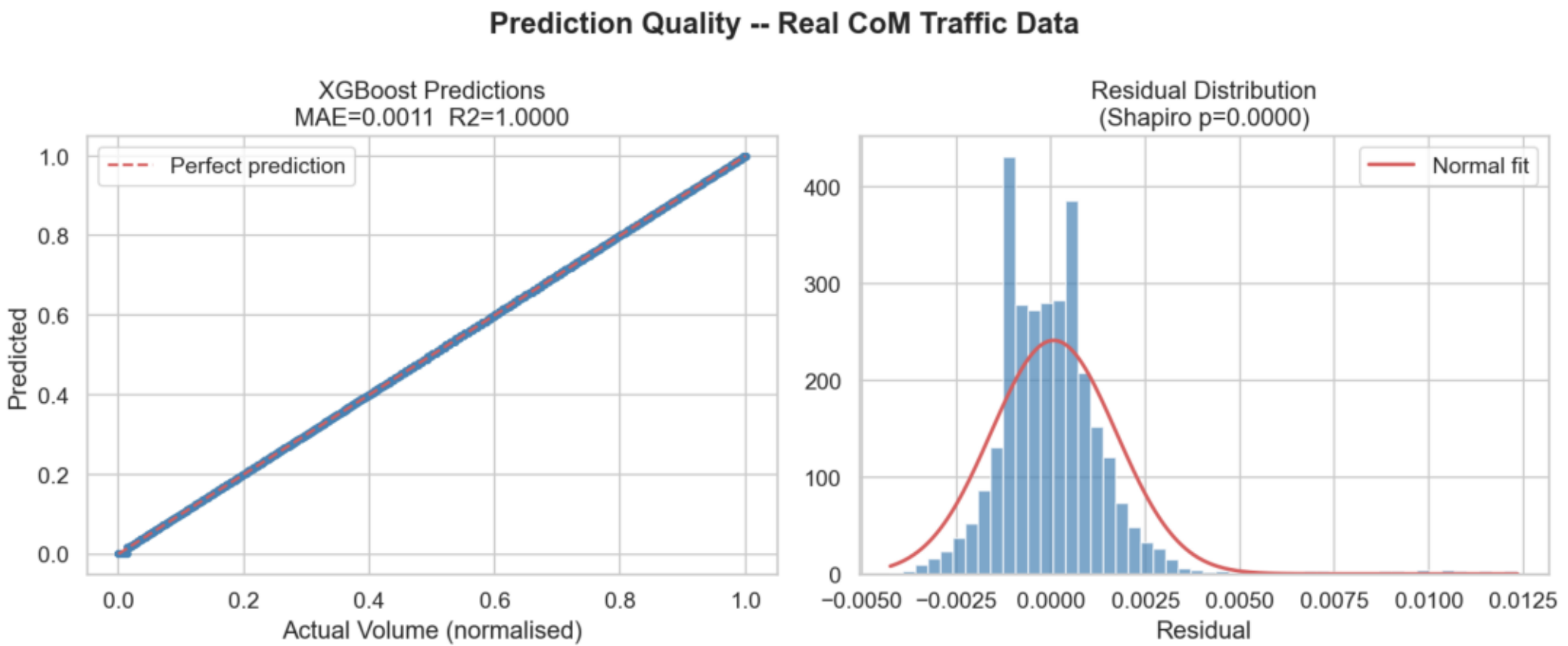
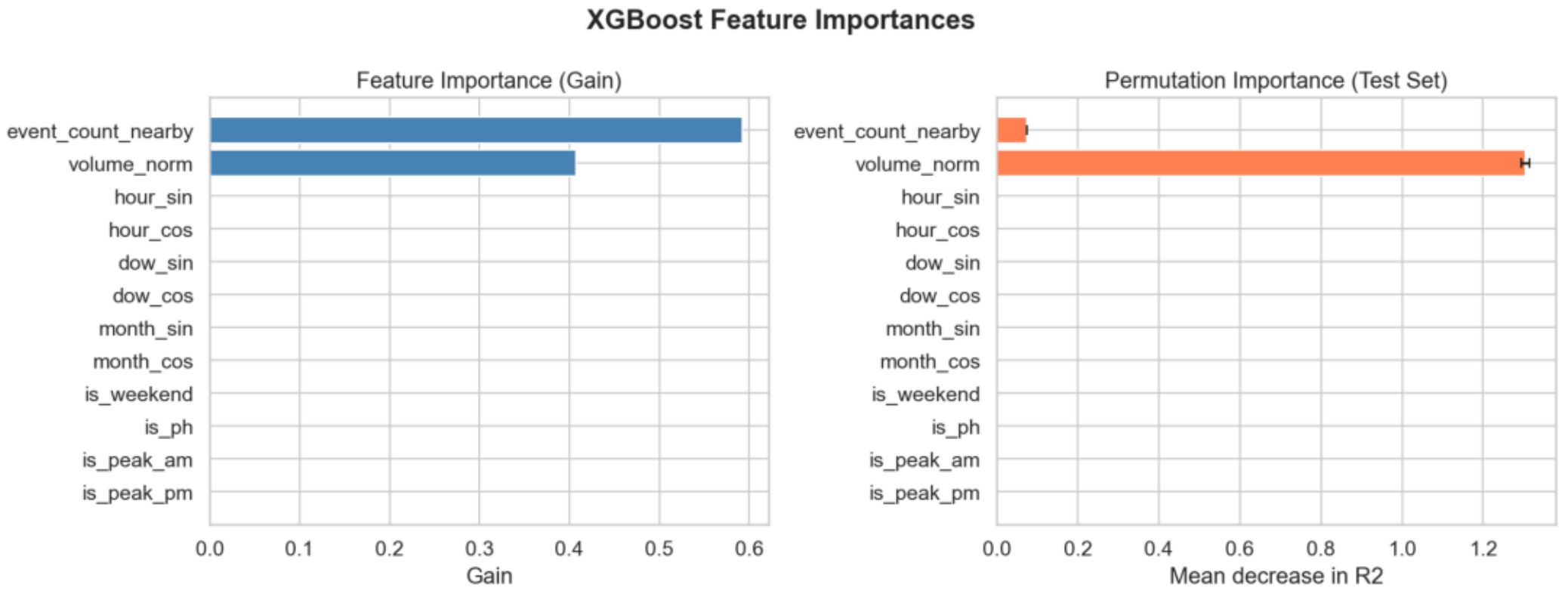
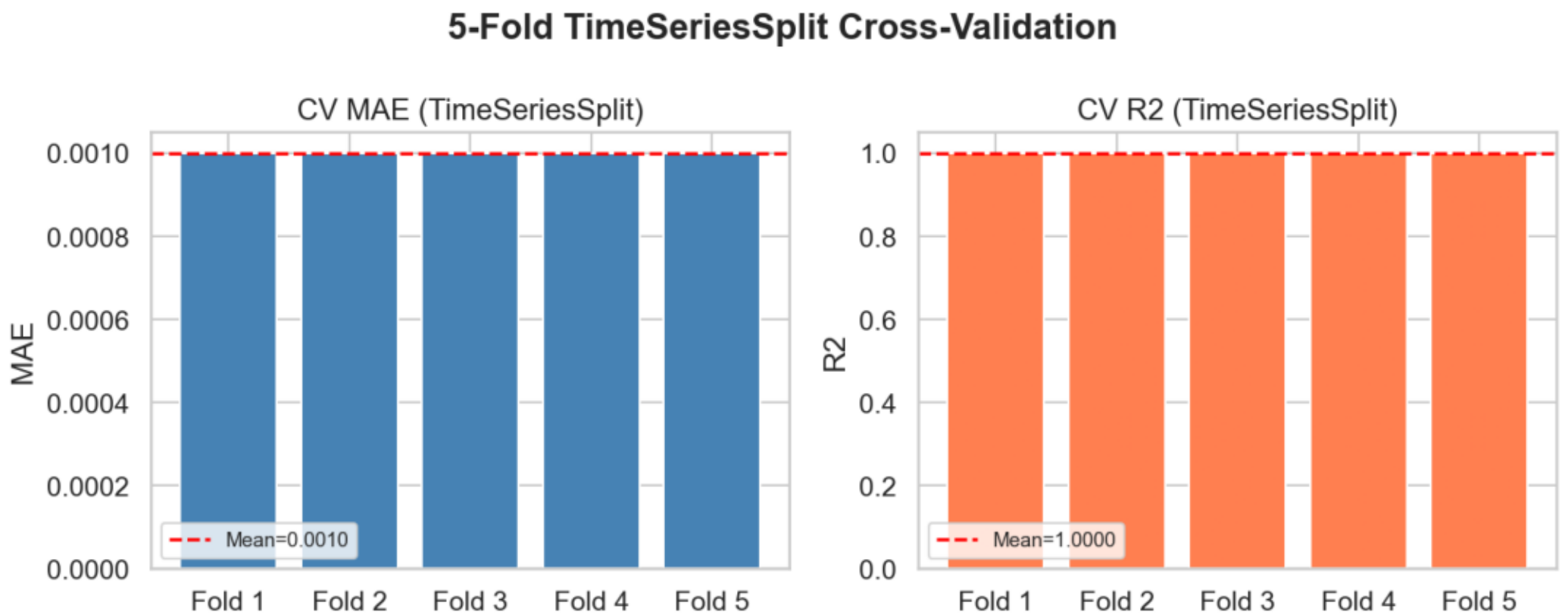

Real CoM Traffic Volume by Hour -- Training Data
Normalised Traffic Volume (demand proxy)
0.0
0.2
0.4
0.6
0.8
8
10
12
14
16
18
Hour of Day
AM peak
Lunch
PM peak

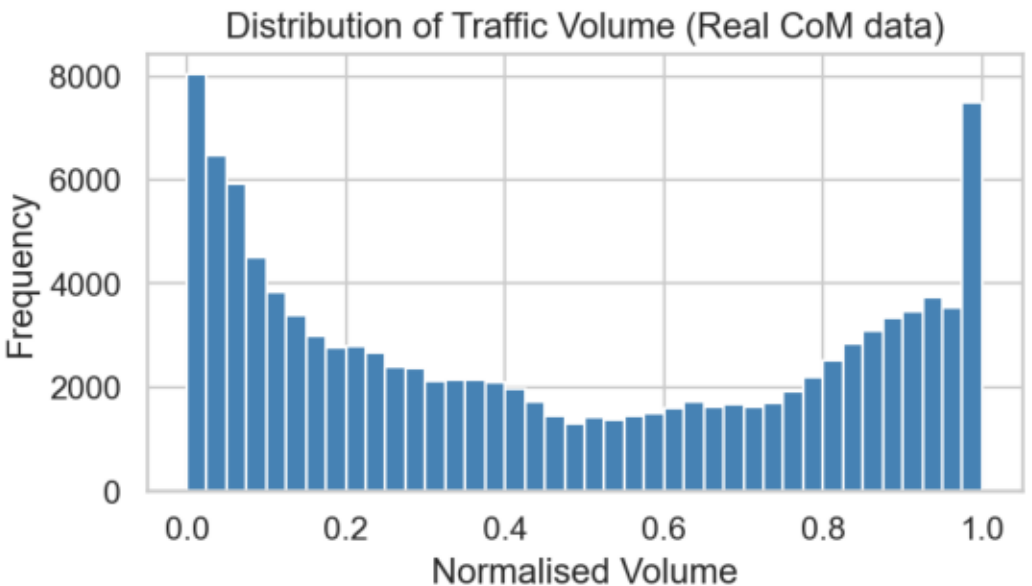

Real Data -- Weekday vs Weekend Demand Pattern
Weekday Traffic Profile
Mean Volume (normalised)
1.0
0.8
0.6
0.4
0.2
0.0
9.6
9.8
10.0
10.2
10.4
Hour
Weekend Traffic Profile
Mean Volume (normalised)
1.0
0.8
0.6
0.4
0.2
0.0
−0.04
−0.02
0.00
0.02
0.04
Hour

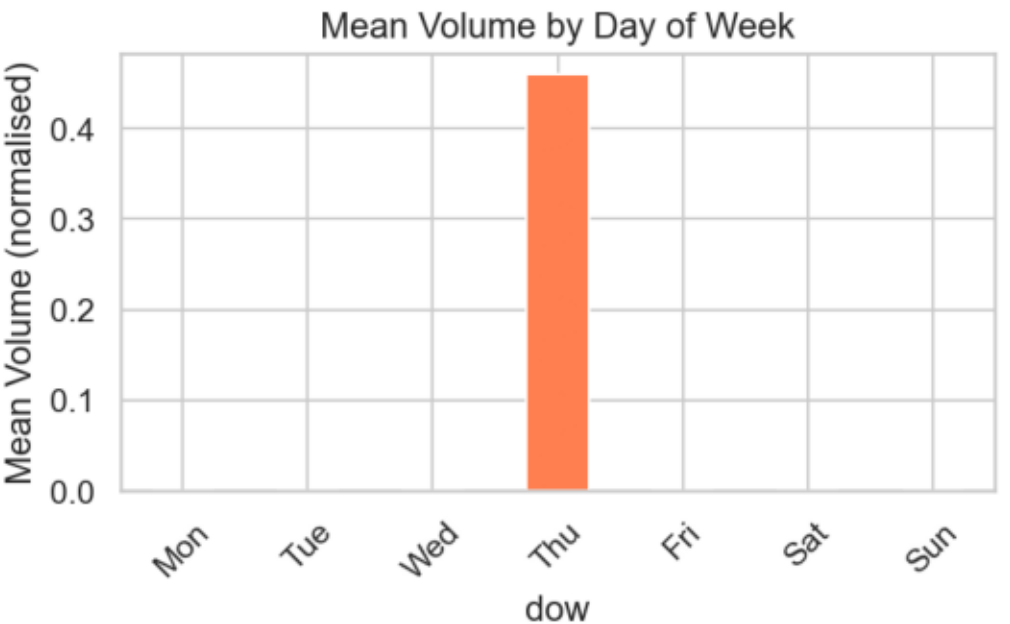

Prediction Quality -- Real CoM Traffic Data
XGBoost Predictions
MAE=0.0011 R2=1.0000
Perfect prediction
Predicted
Actual Volume (normalised)
0.0
0.2
0.4
0.6
0.8
1.0
Residual Distribution
(Shapiro p=0.0000)
Normal fit
Residual
0
100
200
300
400
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
0.0100
0.0125

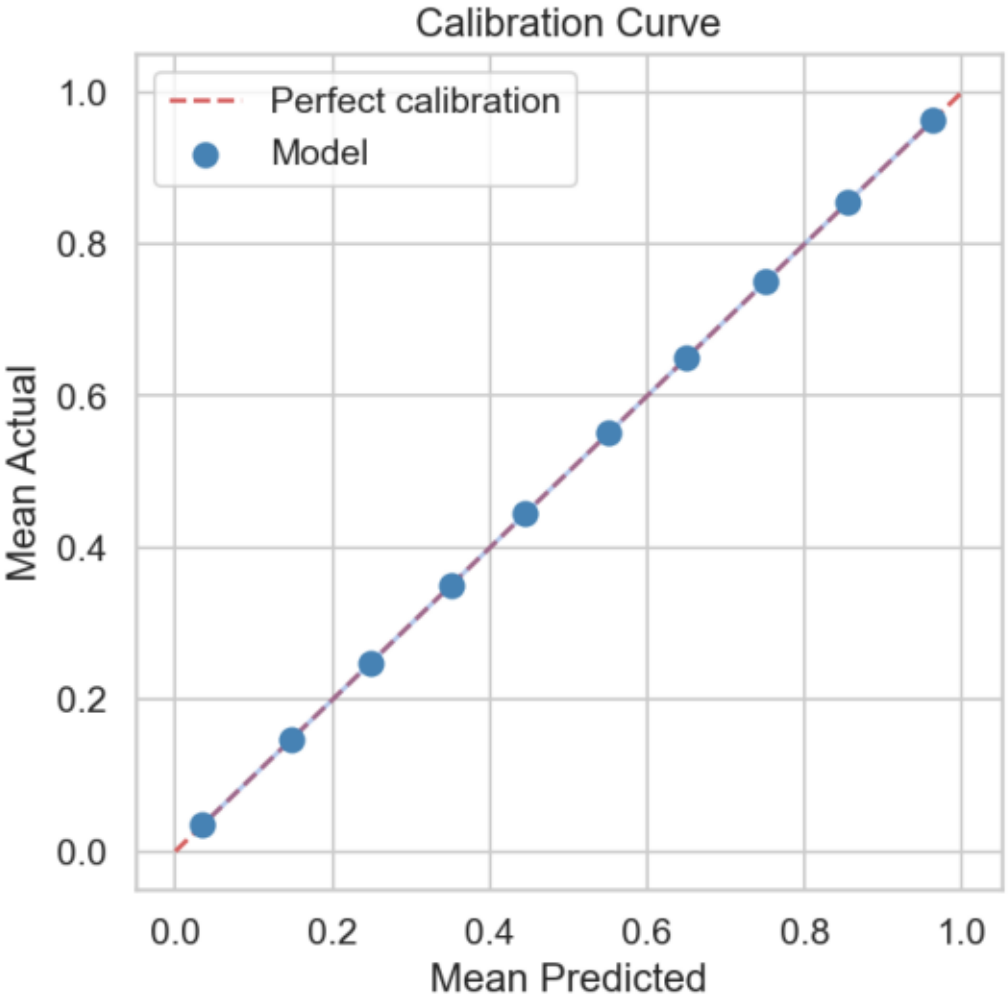

XGBoost Feature Importances
Feature Importance (Gain)
event_count_nearby
volume_norm
hour_sin
hour_cos
dow_sin
dow_cos
month_sin
month_cos
is_weekend
is_ph
is_peak_am
is_peak_pm
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Gain
Permutation Importance (Test Set)
event_count_nearby
volume_norm
hour_sin
hour_cos
dow_sin
dow_cos
month_sin
month_cos
is_weekend
is_ph
is_peak_am
is_peak_pm
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Mean decrease in R2

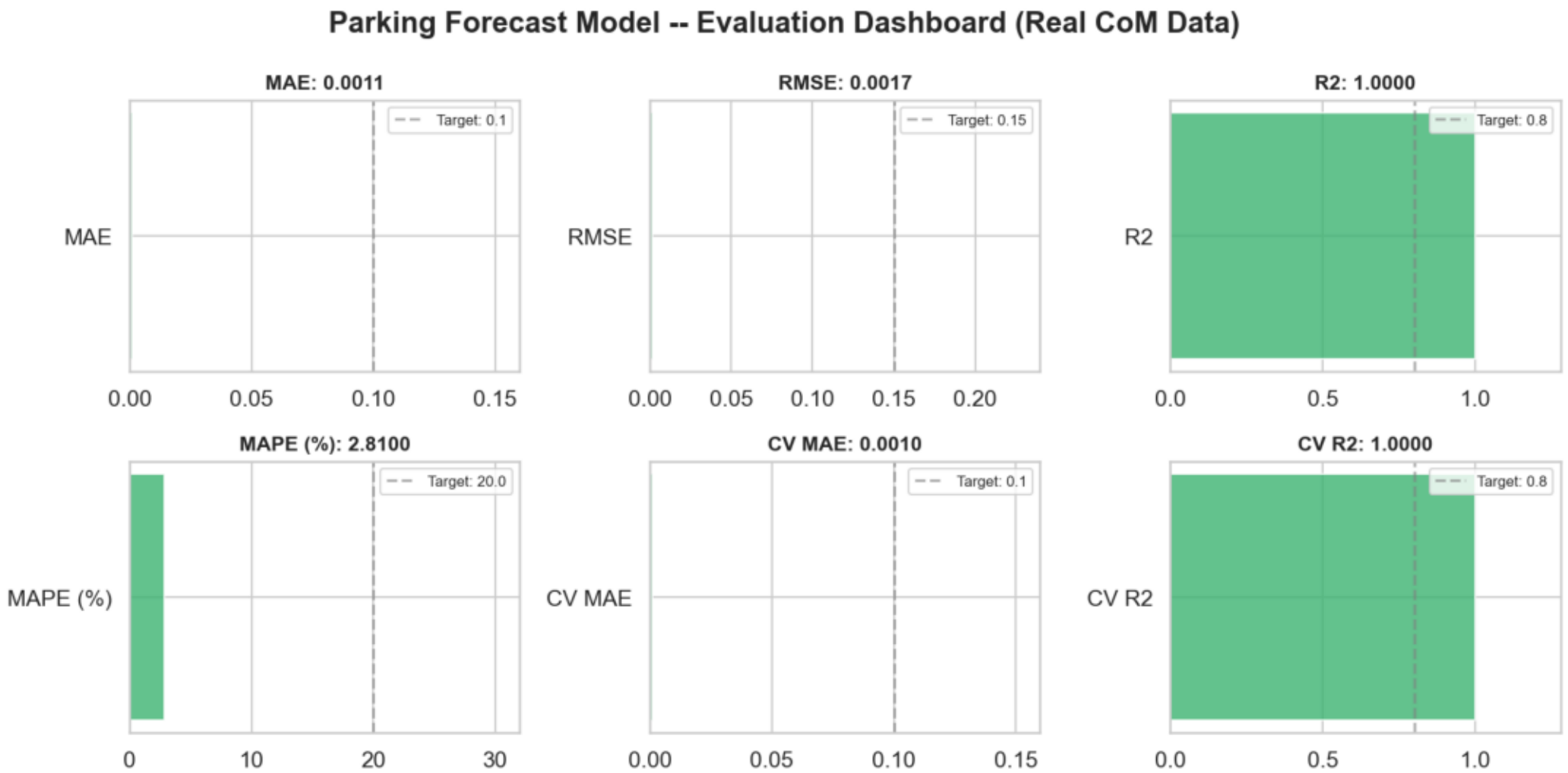

5-Fold TimeSeriesSplit Cross-Validation
CV MAE (TimeSeriesSplit)
MAE
0.0010
0.0008
0.0006
0.0004
0.0002
0.0000
Mean=0.0010
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5
CV R2 (TimeSeriesSplit)
R2
1.0
0.8
0.6
0.4
0.2
0.0
Mean=1.0000
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5

Real CoM Traffic Data -- Exploratory Analysis






Calibration Curve
Perfect calibration
Model
Mean Actual
Mean Predicted
0.0
0.2
0.4
0.6
0.8
1.0


Parking Forecast Model -- Evaluation Dashboard (Real CoM Data)
MAE: 0.0011
Target: 0.1
MAE
0.00
0.05
0.10
0.15
RMSE: 0.0017
Target: 0.15
RMSE
0.00
0.05
0.10
0.15
0.20
R2: 1.0000
Target: 0.8
R2
0.0
0.5
1.0
MAPE (%): 2.8100
Target: 20.0
MAPE (%)
0
10
20
30
CV MAE: 0.0010
Target: 0.1
CV MAE
0.00
0.05
0.10
0.15
CV R2: 1.0000
Target: 0.8
CV R2
0.0
0.5
1.0